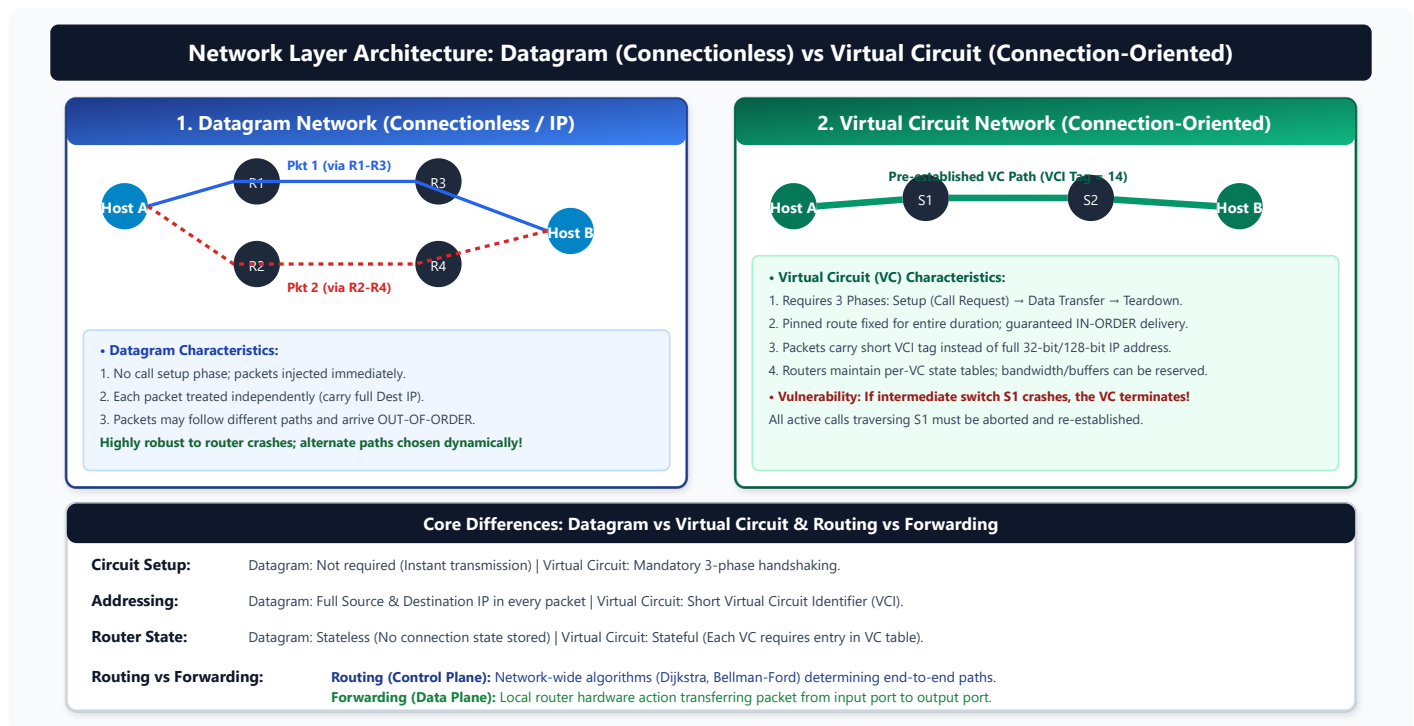


Module 01: Network Layer Foundations, Packet Switching & Virtual Circuits vs Datagram

Unit 3: Network Layer

Packet Switching & Switching Paradigms

End-to-end host delivery mechanics, Store-and-Forward packet switching, complete architectural comparison between Datagram and Virtual Circuit networks, and the Control Plane vs Data Plane separation.



1. Network Layer Responsibilities

The network layer provides end-to-end host-to-host packet delivery across heterogeneous internetworks. Key responsibilities include logical addressing (IP), path routing (control plane), local hardware forwarding (data plane), packetization, and fragmentation.

2. Datagram vs Virtual Circuit Switching

Architectural Philosophy

Datagram: Stateless routers, full IP header per packet, dynamic rerouting upon failure.

Virtual Circuit: Stateful switches, short VCI labels, pre-reserved bandwidth, single-router failure aborts circuit.